

Office of Cannabis Management**Required Testing of Each Lot of Adult Use Cannabis and Medical Cannabis
Product**

This document lists the permitted analytes and sets testing limits for contaminants as directed in 9 New York Codes, Rules and Regulations (NYCRR) Part 130 and Cannabis Law. Testing for each contaminant is required for all final adult- use and medical cannabis product types unless expressly noted. Additional analytes may be added. These limits may be modified where it is in the best interest of public health and safety.

Regulatory Authority

Pursuant to the authority vested in the Cannabis Control Board (Board) by Sections 13, 43, 89, 105 and 129 of the Cannabis Law, 9 NYCRR Part 130, specifically Section 130.22 Testing of Cannabis Product and Medical Cannabis:

- (a) testing of the phytocannabinoid profile in cannabis products and/or medical cannabis shall include, at a minimum, the analyte or groups of analytes specified under this Part; and
- (b) testing for contaminants in cannabis product or medical cannabis shall include, but not be limited to, microorganisms, foreign material, metals, microbial impurities, moisture content and water activity, mycotoxins, pesticides, residual solvents, terpenoids, and any other analyte or group of analytes determined by the Office, consistent with the acceptable limits determined by the Office for each of the foregoing.

Revision Record

Date	Summary of Change
<i>NYS OCM Adult-Use and Medical Cannabis Testing Program</i>	
February 2026	Aligned analyte or group of analytes with Metrc Product Category Chart for physical chemistry testing.
December 2025	- Amended limits for filth & foreign material to be consistent with testing standards to report a quantified result for mammalian excreta and a quantified result for foreign material. - Aligned analyte or group of analytes with Metrc Product Category Chart. - Aligned reporting of results for phytocannabinoid profile with PML regulations and Metrc.
September 2025	Amended Table 3 to address applicable TYM and TAB limits for mixed matrix cannabis products providing further clarification for laboratories and licensees.
August 2025	- Clarified what limits are applicable to mixed matrix cannabis, including infused and enhanced flower products. - Aligned applicability to Cannabis Analyte Testing Specifications flow chart.
May 2025	- Clarified what tests are applicable to solventless and solvent-based hash and cannabis products coated with concentrates. - Clarified max terpene limit for inhalables with 15% variance.
February 2025	- Clarified when physical chemistries are required on cannabis products. For example, pre-rolls are not required to be tested for filth & foreign material. - Clarified when TYM and TVAB are to be tested on enhanced flower products and resins such as rosins, kief, and hash. - Clarified when residual solvents are to be tested. - Clarified unit of measure to use for cannabinoids.
July 2024	- Updated Cadmium limit as USP was updated. - Amended adult-use reference in Table 1 from guidance to regulation.
March 2024	Updated to include testing for the residual solvent, 1,1,1,2-Tetrafluoroethane (HFC-134a). Limit was set at 1000 ppm.
January 2024	- Additional (CAS) registry numbers were added to some of the terpenes based on individual or mixed isomer standards available on the market. - Clarified how terpene results are to be reported.
November 2023	- CAS registry numbers were added next to each terpene. - Clarified terpene limit of more than 10% is applicable to only vaporized or inhaled cannabis products.
October 2023	- Clarified the limits with either a < or = sign, or both, and amended limits to significant figures based on accuracy of method. - Included criteria for labeled potency variance and potency homogeneity. - Included a required list of terpenes to be tested for and the limit as a % by weight.

	• Included reporting criteria for pesticides and residual solvents that are identified as “TIC” on reports. • Included clarification as to which types of products water activity is applicable to. • Included clarification as to which types of products moisture content is applicable to. • Added clarification for metals limit for topicals. • For adult-use products, clarified vape and flower are to be reported as %cannabinoid and total THC and total CBD per package rather than per serving. • Amended the limit for Nickel in AU flower products only.
January 2023	• Limit for Total Aerobic Microbial Count and Total Yeast/Mold Count in adult-use products containing unextracted cannabis was changed to “Report Results only,” and a footnote was added the table. • For heavy metals, the parenteral column was removed and limits for Cu, Cr and Ni were changed back to those listed in the April 2022 document.
August 2022	• Regulatory reference (Part 130) was included. • Phytocannabinoid profile testing requirements were included. • Heavy metals table was updated to include a parenteral limit and remove Zn. • Changed the limits for Cu, Cr and Ni as it was stated under the April 2022, testing limits revision. • List of pesticides was included along with their respective limit. • Water activity, moisture content and filth & foreign material were added as regulated analytes with limits. • List of residual solvents was included along with their respective limit.
<i>NYS DOH Medical Cannabis or Marijuana Testing Program (MMP)/NYS Adult-Use and Medical Cannabis Testing Program</i>	
April 2022	• Reference changed from NYS DOH to NYS OCM. • Total Aerobic Microbial Count and Total Yeast/Mold Count were added as microbiological contaminants. • Reduced the number of pathogenic contaminants to Salmonella spp, Shiga toxin-producing Escherichia coli (STEC), and 4 Aspergillus species (flower and inhaled products only). • With April 2022 revision, Zinc (Zn) was removed as a trace / heavy metal of concern. US Pharmacopeia (<232> and <561>) does not consider it a metal of concern requiring a limit.
June 2019	Pyrethrin I & II method was developed.
February 2018	Original required testing limit document was released by NYS DOH MMP. It contained: 10 microbiological contaminants (with pseudomonas for vapor products only), mycotoxins (4 Aflatoxins & 1 Ochratoxin based on tincture/oil matrix), trace elements (9 based on oral and inhalation), and declared pesticides, herbicides, and fungicides (i.e., IBA, Azadirachtin, PBO, and Pyrethrin I & II – <i>method in development</i>).

Phytocannabinoid Profile:

Testing is required on all final cannabis products, which applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

Table 1

Phytocannabinoid Profile	Medical Cannabis Product	Adult-Use Cannabis Product	Lab Result (3 significant figures)
	Limit (+/- %)	Limit (+/- %)	
Tetrahydrocannabinol (THC) as Total THC of:	Pursuant to 9 NYCRR Section 113.12 Labeled potency variance is 90-110%. **	Pursuant to 9 NYCRR Section 123.6 Labeled potency variance is 85-115%.	Report as mass (mg/dose or mg/serving and mg/package) and percentage (% weight)
Δ9-THC;			
Δ8-THC;*			
Δ10-THC (epimers); and*			
the optical isomer of such substances			
Tetrahydrocannabinolic acid (THCA)			
Tetrahydrocannabivarin (THCV)			
Total Cannabidiol (CBD)*			
Cannabinadiolic acid (CBDA)*			
Cannabidivarin (CBDV)			
Cannabinol (CBN)			
Cannabigerol (CBG)			
Cannabichromene (CBC)			

Cannabis Testing Limits

Any Other Marketed Phytocannabinoid			
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* A cannabis laboratory may report individual results for Δ 9-THC Δ 8-THC and Δ 10-THC isomers in adult-use and medical cannabis products. Total THC must always be reported. As defined in Article 1 of the Cannabis Law, Total THC is the sum of the percentage by weight or volume measurement of tetrahydrocannabinolic acid (THCA) multiplied by 0.877, plus the percentage by weight or volume measurement of THC. Total CBD is the sum of the percentage by weight or volume measurement of (CBD + [CBDA x 0.877]). For any other marketed phytocannabinoid, it must be calculated like Total THC and Total CBD using its acid form and 0.877 in the calculation. For bulk flower and plants forms, phytocannabinoids must be corrected for moisture content and reported on a dry weight basis.

** An additional +/-2% lab variability is allowed to be applied for potency acceptance criteria when the total THC concentration or total CBD concentration is less than five (5) milligrams per dose or serving. This does not apply to adult-use products.

Homogeneity Testing for Phytocannabinoid Profile:

Testing is required on final cannabis concentrates and edible products, which applies to the following Metrc Product Category and Lab Test Batch:

1. Concentrate/Extract – Each
2. Infused Edible
3. Infused Liquid Edible – Final Form
4. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
5. Infused Non-Edible – Each
6. Tinctures – Each

It does **not** apply to cannabis flower, including pre-rolls, kief and solventless hash, and mixed matrix cannabis products (e.g., infused or enhanced flower products including pre-rolls) and solvent-based hash, which corresponds to the following Metrc Product Category and Lab Test Batch:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each (Non-Solvent)

Five (5) samples across a cannabis product batch or lot, regardless of batch or lot size, must be tested for potency. Homogeneity must be established among three (3) consecutive batches or lots. Unless otherwise approved by the Office, the concentration of total THC and CBD in milligrams per single dose for any sample of a product submitted for testing must be within twenty-five (25) percent of the mean concentration of total THC and CBD in milligrams per single dose for that submitted lot with the exception that, for products with a specified total THC and CBD concentration less than two (2) milligrams per single dose, the concentration of each sample for that low concentration phytocannabinoid shall be within 0.5 milligrams per dose of the mean concentration.

Microbiological Contaminants:

Testing is required on all final cannabis products, which applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

The regulated microbiological analytes are listed in the tables below. If a non-regulated analyte is detected, it must be identified on the final report. The laboratory is not required to report a result for a non-regulated microorganism in cfu/g or cfu/mL.

Table 2

Microorganism Analyte	Adult-Use Cannabis and Medical Cannabis Product Limit
	(Presence/Absence)
Salmonella species	Absent
Shiga toxin-producing <i>Escherichia coli</i> (O157:H7 plus O26, O111, O103, O121, O45, and O145)	Absent
<i>Aspergillus fumigatus</i> , <i>Aspergillus flavus</i> , <i>Aspergillus niger</i> , <i>Aspergillus terreus</i>	Absent

Table 3

Adult-Use Cannabis Product	Total Viable Aerobic Bacteria Count	Total Yeast and Mold Count
	Report Results Only	
	(in cfu/g or cfu/mL)	(in cfu/g or cfu/mL)
Products containing Unextracted Cannabis ^{*, **}	Report Results Only	Report Results Only
Products containing Mixed Unextracted and Extracted or Infused Cannabis ^{***}	Report Results Only	Report Results Only
Medical Cannabis Product	Total Viable Aerobic Bacteria Count	Total Yeast and Mold Count
	Limit (≤)	
	(in cfu/g or cfu/mL)	(in cfu/g or cfu/mL)
Products containing Unextracted Cannabis	100,000 or 10 ⁵	10,000 or 10 ⁴
Products containing Mixed Unextracted and Extracted or Infused Cannabis ^{***}	10,000 or 10 ⁴	1,000 or 10 ³
Adult-Use and Medical Cannabis Product	Total Viable Aerobic Bacteria Count	Total Yeast and Mold Count
	Limit (≤)	
	(in cfu/g or cfu/mL)	(in cfu/g or cfu/mL)
Products containing Extracted Cannabis or Infused Cannabis	10,000 or 10 ⁴	1,000 or 10 ³

* Please note, that while Total Viable Aerobic Bacteria Count and Total Yeast and Mold Count tests are required, there will not be a defined limit for unextracted products (e.g., cannabis flower products, including kief) in the adult-use (AU) program. The Office will monitor these laboratory testing results and licensees may be required to conduct further testing where results indicate concerns with product quality or safety.

** Kief and solventless hash are considered unextracted cannabis, and the microbial results are report results only. Refer to Cannabis Analyte Testing Specifications flow chart.

*** Products combining unextracted cannabis with extracted or infused cannabis are considered mixed matrix cannabis products. Examples of mixed matrix cannabis products include pre-rolls coated with diamonds, flower infused with resin, flower dusted with kief, and pre-rolls infused with resin.

Mycotoxins:

Testing is required on all final cannabis products, which applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

Table 4

Mycotoxins	Lab Result	Adult-Use Cannabis and Medical Cannabis Product
		Limit (in µg/g)
Total Aflatoxins (Sum of Aflatoxin B1, Aflatoxin B2, Aflatoxin G1, and Aflatoxin G2, if determined individually)	Report all results in µg/g only	<0.020
Ochratoxin A		<0.020

Trace / Heavy Metals:

Testing is required on all final cannabis products, which applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

Table 5

Trace / Heavy Metal Analyte (as a Total)	Lab Result	Oral *	Inhalation
		Adult-Use Cannabis and Medical Cannabis Product Limit (in µg/g) (≤)	
Antimony (Sb)	Report all results in µg/g only	120	2.00
Arsenic (As)		1.50	0.200
Cadmium (Cd)		0.500	0.200
Chromium (Cr)		1,100	110
Copper (Cu)		300	30.0
Lead (Pb)		0.500	0.500
Mercury (Hg)		3.00	0.100
Nickel (Ni)		20.0	2.00 (5.00**)

* Please note for topicals, the applicable limits are those for oral products.

** The limit of 5.00 µg/g applies to adult-use flower and pre-roll products.

Water Activity:

Testing applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each (Non-Solvent)
7. Infused Edible

A laboratory must analyze flower product forms, including pre-rolls, kief and solventless hash, and mixed matrix cannabis, including infused / enhanced pre-rolls and flower, to determine if the water-activity level is at or below 0.65 units. Results must be reported to two significant figures.

A cannabis laboratory must analyze semi-solid and solid edible infused cannabis products such as fruit chews, chocolates, and lozenges, to determine if the level is at or below 0.85 units. Results must be reported to two significant figures.

Non-edible infused products, oral liquids (e.g., tinctures) and beverages are not subject to water activity testing.

Moisture Content (Loss on Drying):

The flower product forms include pre-rolls, kief and solventless hash, and mixed matrix cannabis, including infused / enhanced pre-rolls and flower.

Testing applies to the following Metro Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each (Non-Solvent)

Flower product samples typically have moisture content in a range of 5.0-15.0%. Percentage must be reported to the nearest tenth. If a sample has a moisture content of greater than 15.0%, the sample is not acceptable.



Filth and Foreign Material:

The flower product forms also include kief but exclude pre-rolls. Testing applies to the following Metrc Product Category:

1. Raw Plant Material – Each
2. Kief – Each

Filth and foreign material include, but is not limited to hair, insects, feces, manufacturing waste, packaging contaminants, and by-products.

Table 6

Filth and foreign material	Limit
Mammalian excreta	Not more than an average of 1mg or more per pound
Foreign material as a total of stems and other foreign matter	Not more than 7% of stems and foreign matter by weight; Individually, not more than an average of 5% of stems 3 mm or more in diameter; and not more than 2% of other foreign matter

Residual Solvents:

Testing is required on all cannabis concentrates, including edibles and beverages, mixed matrix cannabis such as infused or enhanced pre-rolls and infused or enhanced flower, and solventless concentrates. Testing applies to the following Metrc Product Category:

1. Infused Flower – Final Form
2. Infused Pre-Rolls – Each
3. Concentrate/Extract – Each
4. Infused Edible
5. Infused Liquid Edible – Final Form
6. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk *or* Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
7. Infused Non-Edible – Each
8. Tinctures – Each

Testing does **not** apply to the following Metrc Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Concentrate/Extract – Each (Non-Solvent)

All solvents used during the production of adult use and medical cannabis product must be declared for targeted testing. A licensee or registered organization may be required to submit samples for additional testing, including testing for analytes that are not listed below.

Please note that the lack of individual and mixed standards may not be commercially available for all the residual solvents listed below. If this is the case, it will make quantitation of individual peaks challenging and some isomers may co-elute. Where applicable, results should be reported as “tentatively identified, but not quantitatively confirmed.” Place “TIC” as the result and define the term in the footnote of the report.

Table 7

Residual Solvent Analyte	CAS Registry Number	Class # ¹	Adult-Use Cannabis and Medical Cannabis Product Limit (≤)
			(in ug/g = ppm)
1,2-Dichloroethane (Ethylene dichloride, Ethylene chloride)	107-06-2	1	5.00
2-Propanol (Isopropanol, Isopropyl alcohol)	67-63-0	3	5000
Acetone (2-Propanone)	67-64-1	3	5000
Acetonitrile	75-05-8	2	410
Butanes, Total	106-97-8, 75-28-5	n/a	5000
Benzene	71-43-2	1	2.00
Chloroform	67-66-3	2	60.0
Dichloromethane (Methylene chloride)	75-09-2	2	600
Dimethyl sulfoxide (DMSO)	67-68-5	3	5000
Ethanol (Ethyl alcohol)	64-17-5	3	5000
Ethyl acetate (Acetic acid ethyl ester)	141-78-6	3	5000
Ethyl ether (Diethyl ether, 1,1'-Oxybisethane)	60-29-7	3	5000
Heptane (n-Heptane)	142-82-5	3	5000
Hexanes, Total	110-54-3, 107-83-5, 96-14-0, 75-83-2, 79-29-8	2	290
Methanol (Methyl alcohol)	67-56-1	2	3000
Pentanes, Total	109-66-0, 78-78-4, 463-82-1	3	5000
Propane	74-98-6	n/a	5000
Tetrafluoroethane (1,1,1,2-) (HFC-134a)	811-97-2	n/a	1000
Toluene (Methylbenzene)	108-88-3	2	890
Trichloroethane (1,1,1-)	71-55-6	1	1500
Xylenes, Total (ortho-, meta-, para-)	95-47-6, 108-38-3, 106-42-3	2	2170

¹ Class 1 are solvents to be avoided. Class 2 are solvents to be limited. Class 3 are solvents with low toxic potential.

Pesticides, including Growth Regulators and Myclobutanil:

Testing is required on all final cannabis products, including beverages. Testing applies to the following Metrc Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk or Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

Any pesticide, including growth regulators and myclobutanil, used during production of an adult use and medical cannabis product must be declared for targeted testing. Targeted testing will be supplemented by the testing of the following pesticide contaminants and their limits. A licensee or registered organization may also be required to submit samples for additional testing, including testing for analytes that are not listed below.

Please note that the lack of individual and mixed standards may not be commercially available for all the pesticides listed below. If this is the case, it will make quantitation of individual peaks challenging and some isomers may co-elute. Where applicable, the results should be reported as *“tentatively identified, but not quantitatively confirmed.”* Place “TIC” as the result for that pesticide and define the term in the footnote of the report.

Table 8

Pesticide Analyte	CAS Registry Number	Adult-Use Cannabis and Medical Cannabis Product Limit ($\leq$)
		(in ppm = 1 μ g/g)
Abamectin	71751-41-2	0.500
Acephate	30560-19-1	0.400
Acequinocyl	57960-19-7	2.00
Acetamiprid	135410-20-7	0.200
Aldicarb	116-06-3	0.400

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Azadirachtin	11141-17-6	1.00
Azoxystrobin	131860-33-8	0.200
Bifenazate	149877-41-8	0.200
Bifenthrin	82657-04-3	0.200
Boscalid	188425-85-6	0.400
Captan	133-06-2	1.00
Carbaryl	63-25-2	0.200
Carbofuran	1563-66-2	0.200
Chlorantraniliprole	500008-45-7	0.200
Chlordane	57-74-9	1.00
Chlorfenapyr	122453-73-0	1.00
Chlormequat chloride	999-81-5	1.00
Chlorpyrifos	2921-88-2	0.200
Clofentezine	74115-24-5	0.200
Coumaphos	56-72-4	1.00
Cyfluthrin	68359-37-5	1.00
Cypermethrin	52315-07-8	1.00
Daminozide	1596-84-5	1.00
Diazinon	333-41-5	0.200
Dichlorvos (DDVP)	62-73-7	1.00
Dimethoate	60-51-5	0.200
Dimethomorph	110488-70-5	1.00
Ethoprop(hos)	13194-48-4	0.200
Etofenprox	80844-07-1	0.400
Etoazole	153233-91-1	0.200
Fenhexamid	126833-17-8	1.00
Fenoxycarb	72490-01-8	0.200
Fenpyroximate	111812-58-9	0.400
Fipronil	120068-37-3	0.400
Flonicamid	158062-67-0	1.00
Fludioxonil	131341-86-1	0.400
Hexythiazox	78587-05-0	1.00
Imazalil	35554-44-0	0.200
Imidacloprid	138261-41-3	0.400
Indole-3-butyric acid	133-32-4	1.00
Kresoxim-methyl	143390-89-0	0.400
Malathion	121-75-5	0.200
Metalaxyl	57837-19-1	0.200
Methiocarb	2032-65-7	0.200
Methomyl	16752-77-5	0.400
Methyl parathion	298-00-0	0.200

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Mevinphos	7786-34-7	1.00
MGK-264	113-48-4	0.200
Myclobutanil	88671-89-0	0.200
Naled	300-76-5	0.500
Oxamyl	23135-22-0	1.00
Paclobutrazol	76738-62-0	0.400
Pentachloronitrobenzene	82-68-8	1.00
Permethrins, Total	52645-53-1, 54774-45-7, 51877-74-8	0.200
Phosmet	732-11-6	0.200
Piperonyl butoxide	51-03-6	2.00
Prallethrin	23031-36-9	0.200
Propiconazole	60207-90-1	0.400
Propoxur	114-26-1	0.200
Pyrethrins ²	8003-34-7	1.00
Pyridaben	96489-71-3	0.200
Spinetoram, Total	187166-15-0, 187166-40-1	1.00
Spinosad, Total	131929-60-7, 131929-63-0	0.200
Spiromesifen	283594-90-1	0.200
Spirotetramat	203313-25-1	0.200
Spiroxamine	118134-30-8	0.200
Tebuconazole	107534-96-3	0.400
Thiacloprid	111988-49-9	0.200
Thiamethoxam	153719-23-4	0.200
Trifloxystrobin	141517-21-7	0.200

² As the cumulative of Pyrethrin 1, Cinerin 1 and Jasmolin 1 (CAS numbers 121-21-1, 25402-06-6 and 4466-14-2, respectively)

Terpenoids:

Testing is required on all final cannabis products, including beverages. Testing applies to the following Metrc Product Category:

1. Raw Plant Material – Each
2. Raw Pre-Rolls – Each
3. Kief – Each
4. Infused Flower – Final Form
5. Infused Pre-Rolls – Each
6. Concentrate/Extract – Each
7. Concentrate/Extract – Each (Non-Solvent)
8. Infused Edible
9. Infused Liquid Edible – Final Form
10. *Either* Vape Cartridges – Each / Inhalable Concentrate – Bulk *or* Vape Cartridge – Final Form / Inhalable Concentrate – Final Form
11. Infused Non-Edible – Each
12. Tinctures – Each

Terpenoids or terpenes are primarily responsible for the aroma of cannabis. All cannabis products must be tested for terpene content.

Vaporized or inhaled cannabis products cannot exceed more than ten (10) percent total terpenes with a plus 15% variance (i.e., 11.5%), which applies to the following Metrc Product Category and Lab Test Batch:

1. AU and Medical (Bud/Flower – Each; Shake/Trim – Each) Raw Plant Material – Each,
2. AU and Medical (Concentrate/Extract – Each) Concentrate/Extract – Final Form,
3. AU and Medical (Concentrate/Extract – Each) Non-Solvent Concentrate – Final Form,
4. AU and Medical (Infused Flower – Each) Infused Flower – Final Form,
5. AU and Medical (Infused Pre-Roll – Each) Infused Prerolls – Final Form,
6. AU and Medical (*either* Vape Cartridge – Each *or* Vape Cartridge – Final Form) *either* Inhalable Concentrate – Bulk *or* Inhalable Concentrate – Final Form
7. AU and Medical (Kief – Each) Kief – Final Form,
8. AU and Medical (Raw Pre-Roll – Each) Raw Prerolls - Final Form

The terpenoids shall be reported in a percentage (by weight). Percentage reported must be based on the readability of the analytical balance used to prepare samples for terpene analysis. For example, a sample was weighed on a balance that reads to 0.001 g; therefore, the result must be reported to 0.001%.

The terpenes listed below are included in current certified reference material (CRM). At a minimum, the following terpenes shall be tested.

Table 9

Terpenoid or Terpene (% by weight)	CAS Registry Number	Additional CAS Registry Number and/or Terpene Name
Beta (β)-myrcene	123-35-3	
Beta (β)-caryophyllene	87-44-5	Trans-caryophyllene
Limonene	5989-27-5	(R)-(+)-Limonene; d-Limonene
Alpha (α)-pinene	80-56-8	
Alpha (α)-humulene	6753-98-6	
Linalool	78-70-6	
Beta (β)-pinene	127-91-3	(-)-β-Pinene as 18172-67-3
Terpinolene	586-62-9	
Ocimene	13877-91-3	Ocimene (mix of isomers); Beta (β)-ocimene
Alpha (α)-bisabolol	23089-26-1	(-)-α-Bisabolol
Caryophyllene-oxide	1139-30-6	(-)-Caryophyllene oxide
Geraniol	106-24-1	
Camphene	79-92-5	
Guaiol	489-86-1	(-)-Guaiol
Alpha (α)-terpinene	99-86-5	
Terpineol (mix of isomers)	8000-41-7	alpha (α)-Terpineol [or (-)-alpha-Terpineol] as 98-55-5
Fenchol	2217-02-9	(1R)-endo-(+)-Fenchyl alcohol
Valencene	4630-07-3	(+)-Valencene
Alpha (α)-phellandrene or p-Mentha-1,5-diene	99-83-2	R-(-)-α-Phellandrene as 4221-98-1
Farnesene (mix of isomers)	502-61-4	Trans-β-Farnesene as 18794-84-8